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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

LAB WEEK - 3

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Introduction

The **Community Micro-Health Monitoring & Alert System** is a dynamic software application where requirements may evolve based on community health conditions, feedback from health workers, and emerging health risks.
Due to this continuously changing nature, the **Agile Model** is chosen for development.

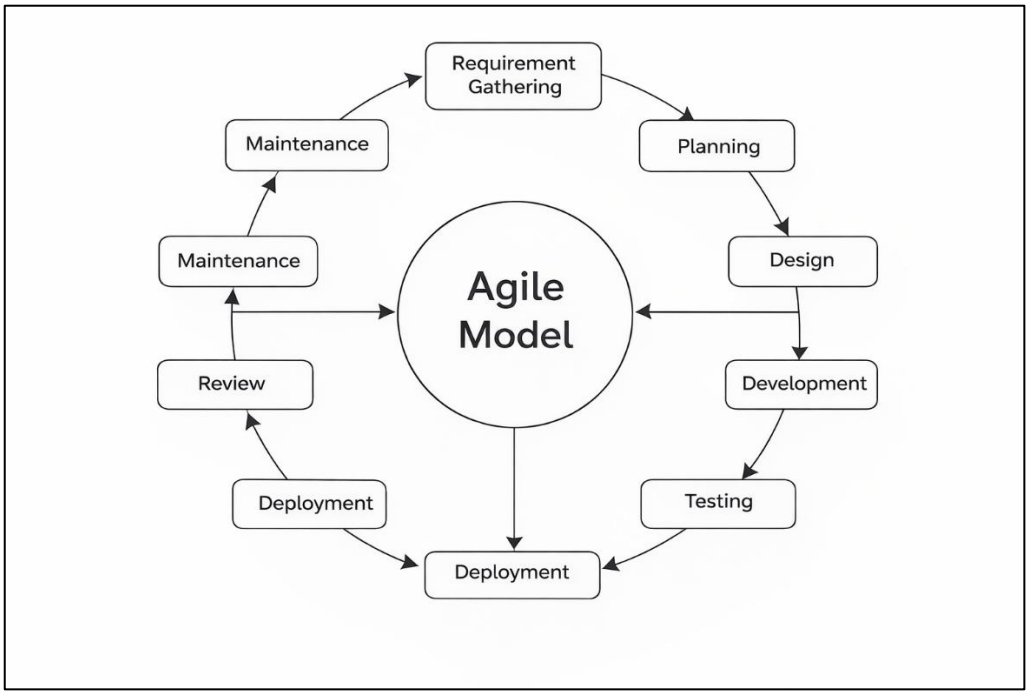
Agile allows the system to be built **incrementally**, enabling early deployment of core features such as health data submission and alert generation, while continuously improving the system through feedback.

Why Agile Model for This Project

The Agile Model is suitable for this project because:

- Health monitoring requirements may change frequently
- Feedback from doctors and health workers is critical
- Early delivery of alert features can save response time
- Continuous improvement improves system reliability

Hence, Agile supports **flexibility, early intervention, and user involvement**, which are essential for community health systems.



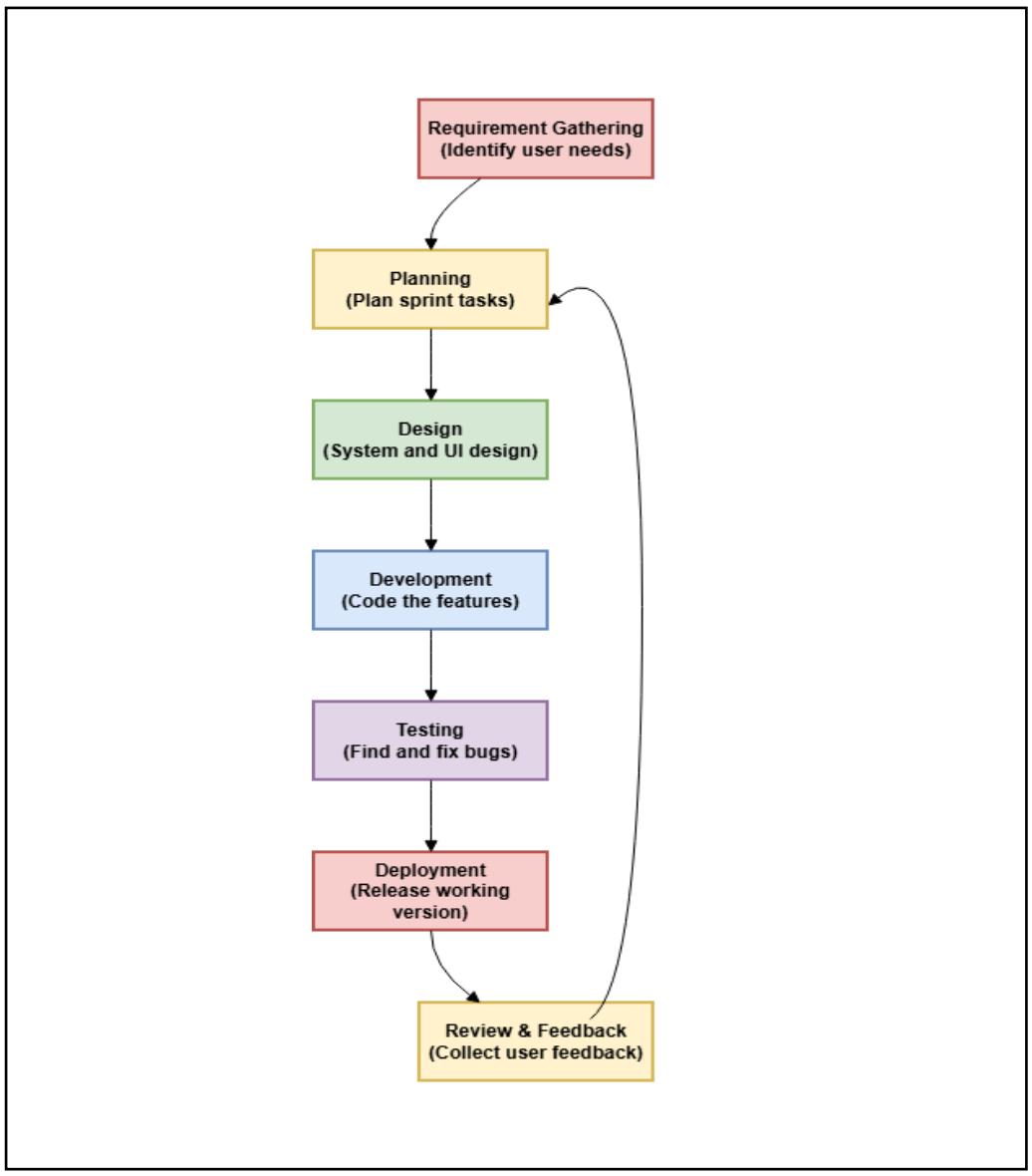
Agile Development Flow for This Project

The project is developed in **short iterations called sprints**, where each sprint delivers a **working part of the system**.

After every sprint:

- The system is reviewed
- Feedback is collected
- Improvements are planned for the next sprint

This ensures **continuous enhancement of health monitoring features**.



Agile Phases Mapped to the Project

Requirement Gathering

Identify health monitoring needs

In this phase:

- Requirements such as symptom logging, risk calculation, and alert generation are identified
- Inputs are collected from citizens, health workers, and doctors

Planning

Plan sprint features

- Features are prioritized (e.g., health data submission first)
- Sprint goals are defined
- Tasks are assigned to the development team

Design

Design system structure

- System architecture is designed
- Database schema for health records is planned
- User interface layouts for citizens and doctors are prepared

Development

Build health monitoring modules

- Health data submission module is implemented
- Risk scoring logic is coded
- Alert notification service is developed

Testing

Verify health data accuracy

- Input validation is tested

- Risk classification is verified
- Alert delivery is tested

Deployment

Release system increment

- The working version is deployed
- Users can start submitting health data
- Alerts are enabled for health workers

Review & Feedback

Improve next sprint

- Feedback from users and health workers is collected
- Issues are identified
- Enhancements are planned for the next sprint

This feedback loop ensures **continuous improvement**.

Sprint-Based Implementation

Sprint	Features Delivered
Sprint 1	User registration and login
Sprint 2	Health data submission
Sprint 3	Risk scoring mechanism
Sprint 4	Alert generation system
Sprint 5	Doctor dashboard and reports

Each sprint delivers a **working increment** of the system.

Sprint-wise Feature Implementation

- **Sprint 1 - User Registration and Login**
  Basic authentication functionality is developed to allow citizens, doctors, and health workers to securely access the system.
- **Sprint 2 - Health Data Submission**
  Citizens are enabled to submit daily health parameters such as symptoms and basic vital information.
- **Sprint 3 - Risk Scoring Mechanism**
  A rule-based logic is implemented to analyze submitted health data and classify cases into low, medium, or high risk.
- **Sprint 4 - Alert Generation System**
  Automatic alerts are generated for health workers and doctors when medium or high-risk cases are detected.
- **Sprint 5 - Doctor Dashboard and Reports**
  A dashboard is developed for doctors to view patient reports, analyze health trends, and monitor high-risk cases.

Agile Roles in This Project

- **Product Owner**
  Defines health monitoring requirements and priorities
- **Scrum Master**
  Ensures smooth Agile execution and sprint completion
- **Development Team**
  Designs, develops, and tests system features

Agile Artifacts Used

Artifact	Usage in Project
Product Backlog	List of all health monitoring features
Sprint Backlog	Features selected for current sprint
Increment	Working system after each sprint

Benefits of Agile for This Project

- Faster detection of health risks
- Continuous system improvement
- Early availability of critical features
- Better collaboration with healthcare staff

Conclusion

The Agile Model enables flexible and incremental development of the **Community Micro-Health Monitoring & Alert System**. By delivering features in small, testable increments and continuously incorporating feedback, Agile ensures that the system remains reliable, responsive, and effective in supporting early health intervention at the community level.